5 Operating Mode

5.1 Overview of Operation Model

The control mode modes of the integrated motor are categorized into CiA402 mode and NiMotion mode. 2002_h : 01_h ($00B1_h$) (control mode option) is used to determine whether the motor is in the CiA402 mode or the NiMotion mode, and the objects are described as follows:

Table 5- 1

Attribution	Description
Index	2002_h
Name	Basic control parameter
Data structure	Record
Data type	-

Sub-index 01h to determine the mode:

Table 5- 2

Sub-Index	01_h ($00B1_h$)
Description	Control mode option
Accessibility	rw
PDO mapping	No
Data type	Unsigned8
Default value	0
Meaning of Value	0x00: CiA402 mode 0x01: NiMotion position mode 0x02: NiMotion speed mode 0x03: NiMotion Torque Mode 0x04: NiMotion open loop mode

After selecting the CiA402 mode, you need to set 6060_h ($03C2_h$) to select the specific motion mode (control mode selection, invalid operation status setting), and the objects are described as follows:

Table 5- 3

Sub-Index	6060_h ($03C2_h$)
Description	Mode Option
Accessibility	rw
PDO mapping	No
Data type	Unsigned8
Default value	0x01
Sub-Index	-
Meaning of Value	0x00: No definition 0x01: Profile Position Mode (PP) 0x02: Velocity Mode (VM)

Meaning of Value	0x03: Profile Velocity Mode (PV) 0x04: Profile Torque Mode (PT) 0x05: No definition 0x06: Home return mode (HM) 0x07: Interpolation mode (IP) 0x08: Cyclic synchronized position mode (CSP) 0x09: Cyclic Synchronized Velocity Mode (CSV) 0x0A: Cyclic synchronous torque mode (CST)

When the servo is enabled and the motor shaft speed is 0, the working mode of the servo driver supports switching between PP and HM, as well as between CSP and HM.

5.2 Position Control Functions

5.2.1 Structure Chart

Target Following Error

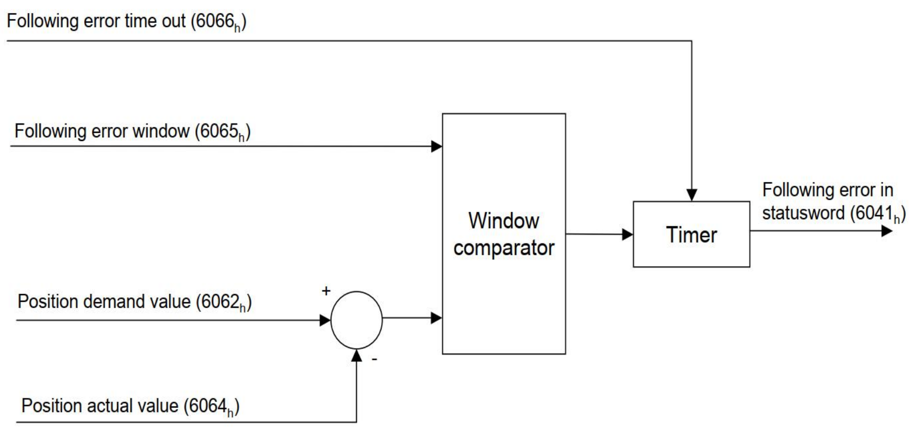

Figure 5- 1

Target Reached

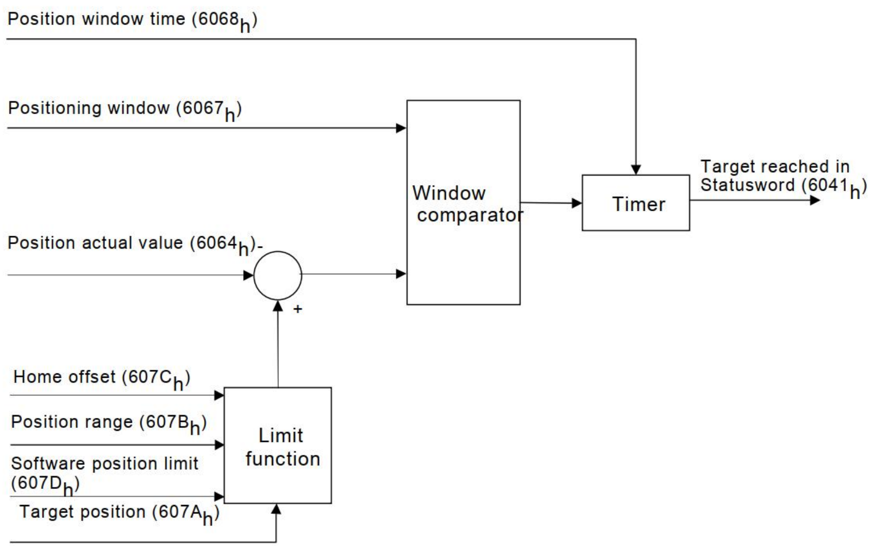

Figure 5- 2

5.2.2 Objects Related

Target following error

Table 5- 4

Object Index	Description
6041_h (0381)$_h$	Status word
6062_h (03C4)$_h$	Current position demand value inside the drive (user unit)
6064_h (03C8)$_h$	Current position actual value feedback (user unit)
6065_h (03CA)$_h$	This object shall indicate the configured range of tolerated position values symmetrically to the position demand value. If the position actual value is out of the following error window, a following error occurs. A following error may occur when a drive is blocked, unreachable profile velocity occurs, or at wrong closed-loop coefficients.
6066_h (03CC)$_h$	Status word

Target Reached

Table 5- 5

Object Index	Description
6041_h (0381_h)	Status word
6064_h ($03C8_h$)	Current position value feedback (user unit)
6067_h ($03CD_h$)	The configured symmetrical range of accepted positions relative to the target position. If the actual value of the position encoder is within the position window, this target position shall be regarded as having been reached.
6068_h ($03CF_h$)	Position window time(unit：ms)
$607A_h$ ($03E7_h$)	Target position(user unit)
$607B_h$	Position range limit(user unit)
$607C_h$ ($03ED_h$)	Difference between the 0-position in the application and the machine home position (user unit)
$607D_h$ ($03EF_h$ and $03F1_h$)	Target position limit (user unit)

5.2.3 Functional Description

Target following error

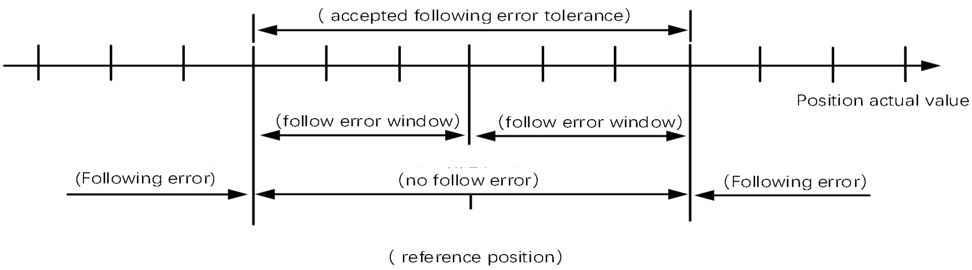

Figure 5- 3

The following error refers to the deviation between the reference position 6062h (03C4h) and the actual position 6064h (03C8h). If the following error value remains greater than the value in the following error window 6065h (03CAh) for the time set by 6066h (03CCh) (as shown in Fig. 6.2, the following error is outside the acceptable following error range), then bit13 (following error) of the status word 6041h (0381h) is set to 1, and the timing diagram is shown below:

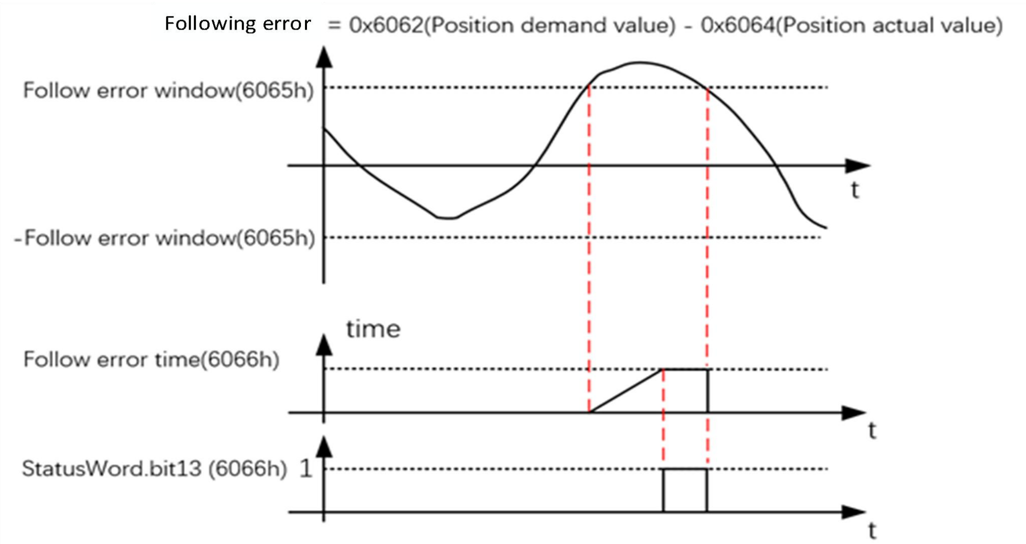

Figure 5- 4

Target Reached

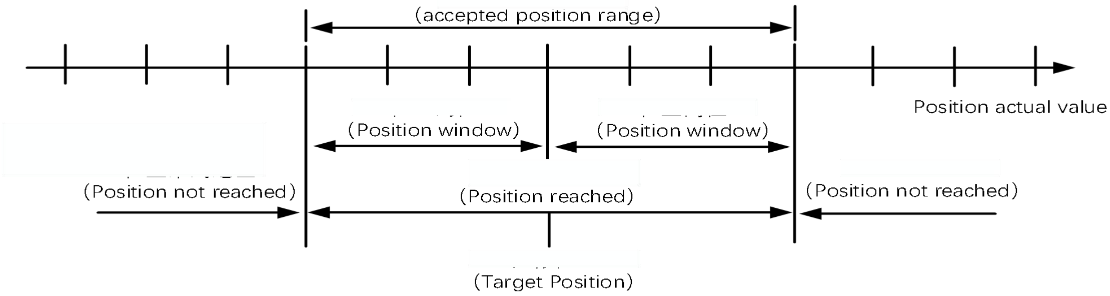

Figure 5- 5 Schematic diagram of the acceptable range of error:

The position difference is the difference between the target position $607A_h$ ($03E7_h$) and the actual position 6064_h ($03C8_h$). If the difference stabilizes within the acceptable position range (as shown above) and reaches the set time 6068_h ($03CF_h$), then bit10 (target reached) of the status word will be set to 1, which means that the target position has been reached. The timing diagram is shown below:

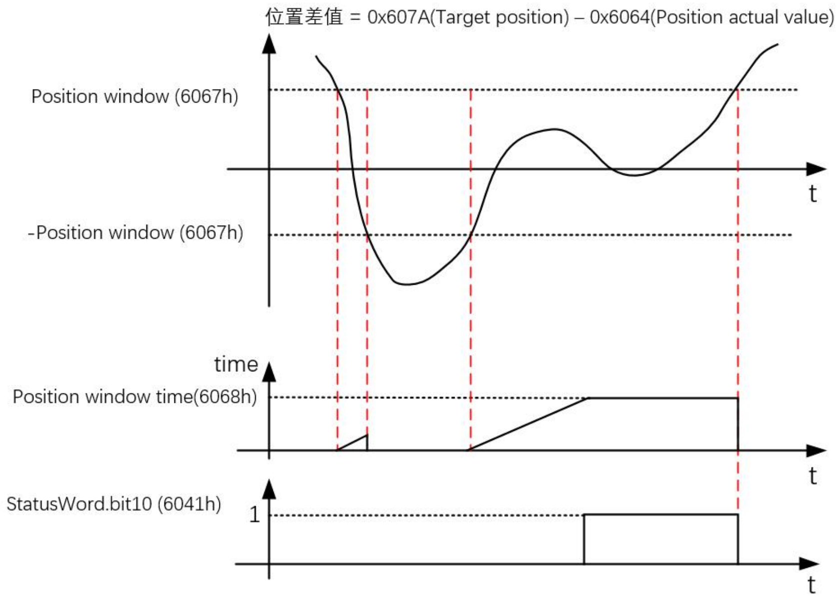

Figure 5- 6

5.3 Contour Position Mode (PP)

This mode is mainly used for point-to-point positioning applications. In this mode, the host computer gives the target position (absolute or relative), the speed of the position curve, acceleration and deceleration and deceleration, the internal trajectory generator will generate the target position curve instruction according to the setting, and the driver will complete the position control, speed control and torque control internally. The overall structure is shown below:

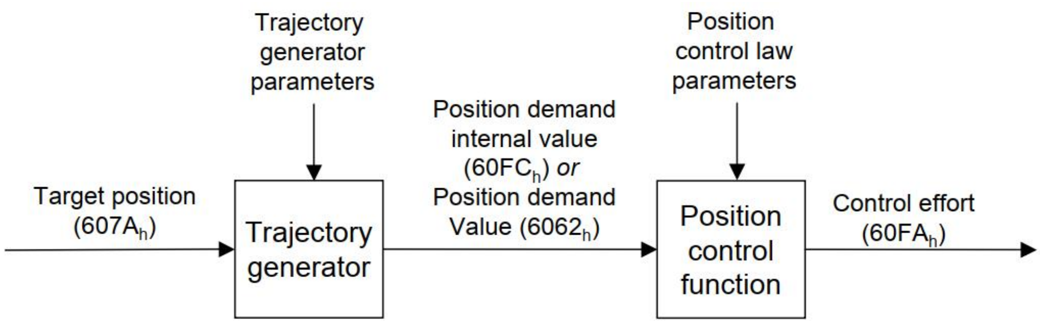

Figure 5- 7

5.3.1 Structure Plan

The following figure defines the detailed structure of the trajectory generator:

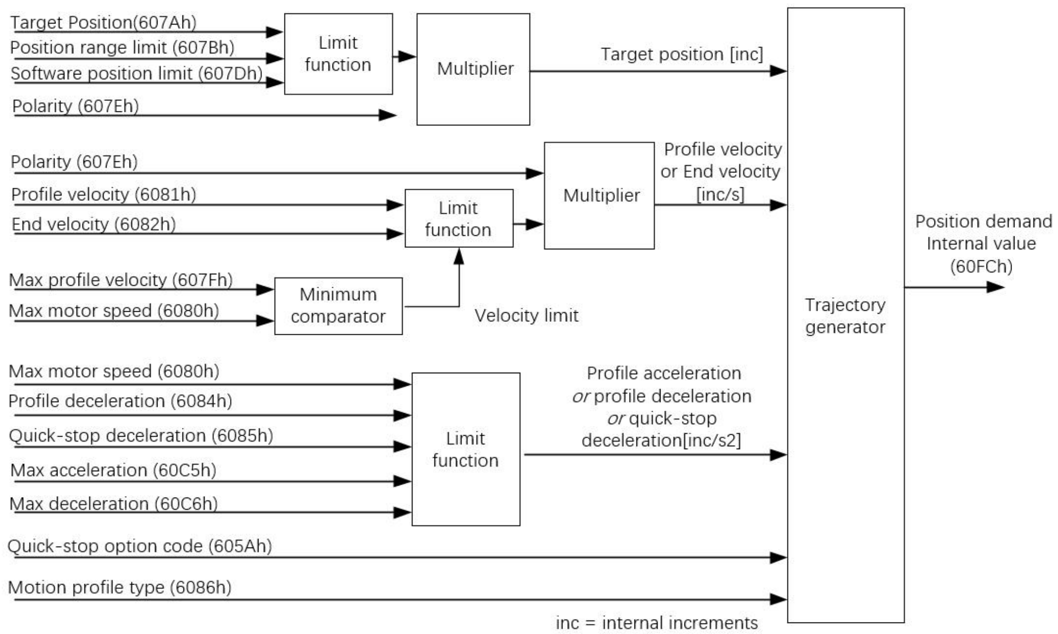

Figure 5- 8

5.3.2 Relevant Object

The following objects require attention in this mode:

Table 5- 6

Object Index	Description
$605D_h$ ($03C0_h$)	Pause mode selection
$605A_h$ ($03BF_h$)	Quick stop mode selection
6062_h ($03C4_h$)	Drive internal current target position command value (user units)
6063_h ($03C6_h$)	Feedback of the current absolute position of the motor (encoder unit)
6064_h ($03C8_h$)	Current position of the motor (user unit)
$607A_h$ ($03E7_h$)	Preset target position (user units)
$607D_h$ ($03EF_h$ and $03F1_h$)	Limitations on target location (user units)
$607E_h$ ($03F3_h$)	Direction of rotation (polarity), see "4.3 $607E_h$ ($03F3_h$): Polarity".
$607F_h$ ($03F4_h$)	Maximum profile speed during operation (user unit/s), which acts as a speed limiter.
6080_h ($03F6_h$)	Maximum motor speed (rpm)
6081_h ($03F8_h$)	The profile velocity (user units/s) of the uniform phase of the run of this segment displacement command, i.e., the velocity at which the end of the acceleration ramp is reached during positioning. The size is limited by $607F_h$ ($03F4_h$).
6082_h ($03FA_h$)	Contour End Velocity, the velocity (in user units/s) at which the target position is reached, the velocity at the end of the ramp, usually this object is set to zero so that the velocity is reduced to exactly 0 when the target position is reached. size is limited by $607F_h$ ($03F4_h$).
6083_h ($03FC_h$)	Contour acceleration during operation (user units/s^2), size limited by $60C5_h$ ($043B_h$).
6084_h ($03FE_h$)	Contour deceleration during the run (user units/s^2), size limited by $60C6_h$ ($043D_h$).
6085_h (0400_h)	Stopping deceleration rate when performing a "quick stop" (user unit/s^2)
6086_h (0402_h)	Selection of the type of planner curve, i.e. slope. If the value is "0", then there will be no limitation of shocks (plus acceleration), i.e. a trapezoidal curve; If the value is "3", the shock (plus acceleration) will be limited by the value of $60A4_h$: 01_h -02_h ($041E_h$ and 0420_h), i.e., S-curve, and only the indexes 01_h and 02_h are used in this device.
$60C5_h$ ($043B_h$)	Maximum acceleration limits (user units/s^2)
$60C6_h$ ($043D_h$)	Maximum deceleration limit (user units/s^2)
$60F2_h$ ($043F_h$)	Positioning Options
$60FC_h$ (0446_h)	Output of the trajectory generator, i.e. internally planned real-time position commands (encoder units)

The following figure illustrates the role of velocity, acceleration, and shock parameters on the running process:

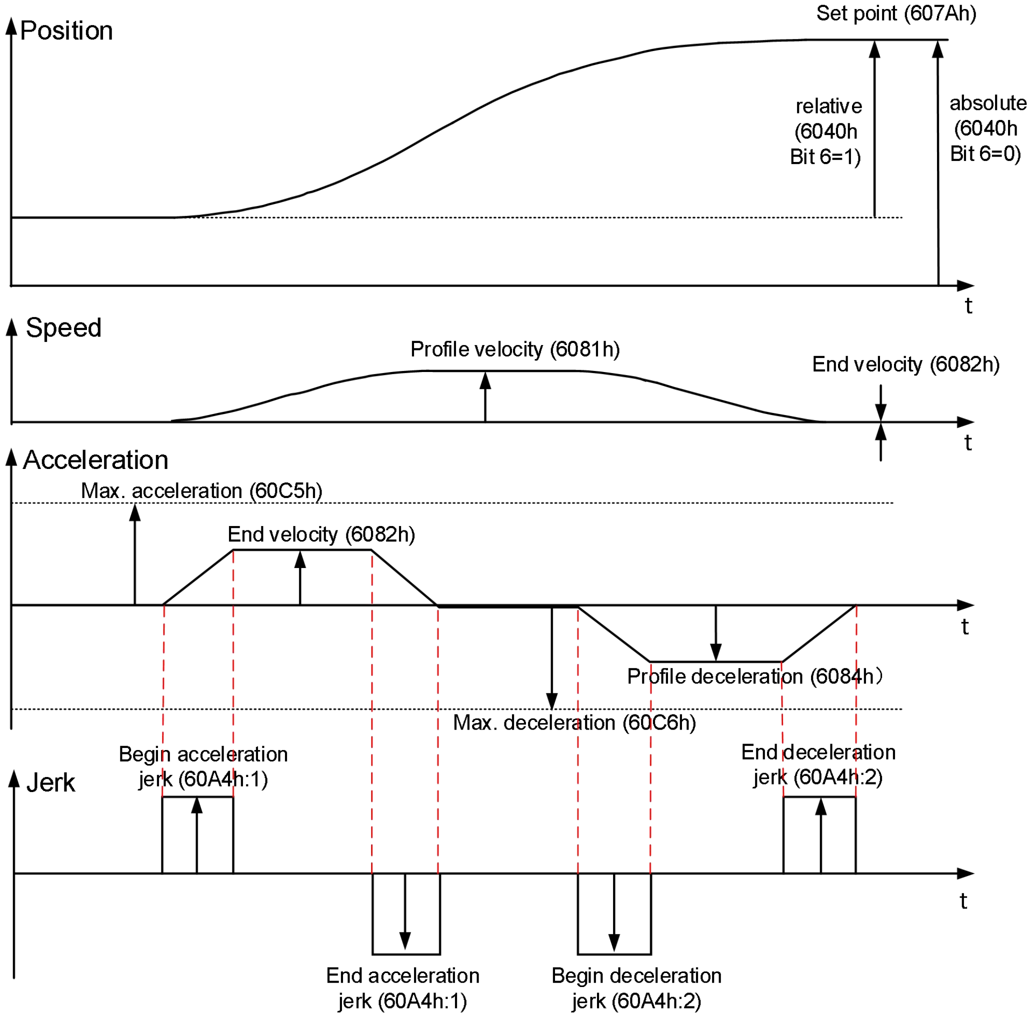

Figure 5- 9

5.3.3 Control Commands and Status Messages

start

To enable this mode, the value "0" must be set in object 2002_h : 01_h ($00B1_h$) and "1" in object 6060_h ($03C2_h$).

control word

15 10	9	8	7	6	5	4	3 0
(see 3.1.2)	Change on set-point	Halt	(see 3.1.2)	abs/rel	Change set immediately	New set-point	(see 3.1.2)

MSB LSB

In profile position mode, the following bits in object 6040_h (0380_h) have a special function

Table 5- 7

Bit 5	Bit 4	Description
0	0->1	Non-immediate updates will complete the running task before starting the next one.
1	0-> 1	Immediate update that will immediately execute the runtime task triggered by bit 4

Table 5- 8

Bit	Value	Description
6	0	The target position is an absolute position command
	1	The target position is a relative position instruction. The target position is a relative position based on the current position, and the reference position depends on bits 0 and 1 of $60F2_h$ ($043F_h$).
8	0	Positioning should be implemented or continued
	1	The motor will decelerate and stop moving, the deceleration depends on the object $605D_h$ ($03C0_h$)

status word

15	14 13	12	11	10	9 0
(see 3.1.3)	Following error	Set-point acknowledge	(see 3.1.3)	reached Target	(see 3.1.3)

MSB LSB

The following bits in object 6041_h (0381_h) have a special function in profile position mode:

Table 5- 9

Bit	Value	Description
10	0	Non-pause state (bit8 = 0 for 6040_h (0380_h)): target not reached Pause state (bit8 = 1 for 6040_h (0380_h)): Motor deceleration
	1	Non-pause state (bit8 = 0 for 6040_h (0380_h)): target reached Pause state (bit8 = 1 for 6040_h (0380_h)): motor speed 0
12	0	Positioning of the previous positioning point is complete, waiting for the

		new set point
	1	The previous locator is still being processed and the new locator will overwrite the old one
13	0	No following error
	1	With following error

5.3.4 Functional Description

The control instruction timing is explained below in terms of immediate and non-immediate updates:

1） Control Instruction Timing - Update Now

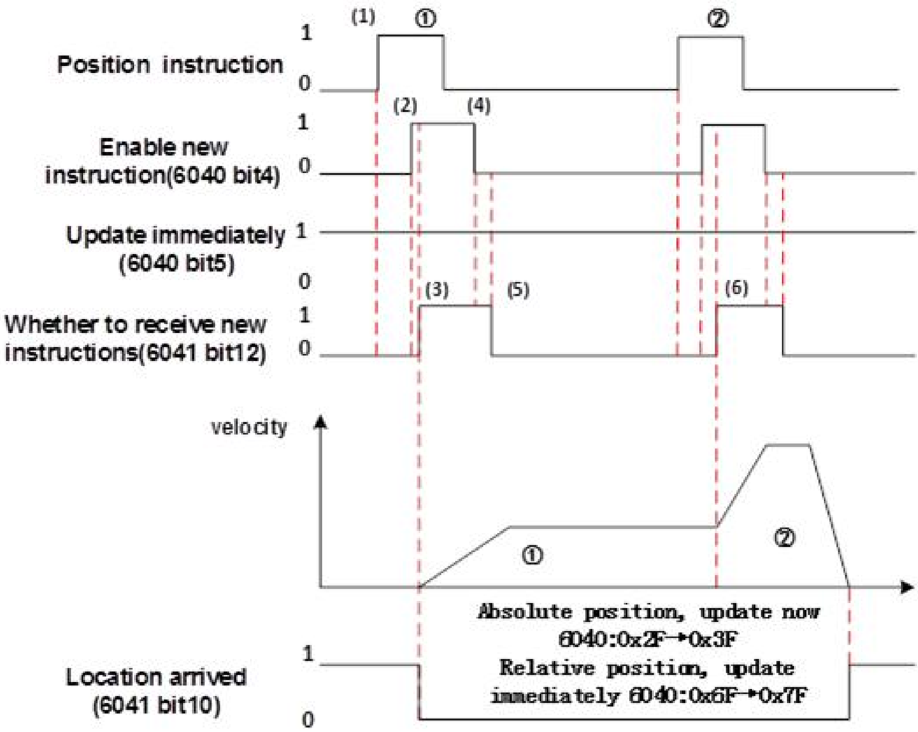

Figure 5- 10

The table below is a paraphrase of some of the labeling that was done during the course of the above diagram:

Table 5- 10

(1)	The upper computer updates the displacement commands (including target displacement $607A_h$ ($03E7_h$), acceleration time 6083_h ($03FC_h$), deceleration time 6084_h ($03FE_h$), profile speed 6081_h ($03F8_h$), etc.)
(2)	Set bit4 of 6040_h (0380_h) from 0 to 1 to indicate a new displacement command from the slave (bit5 of 6040_h (0380_h) is set to 1 for immediate updates)
(3)	After receiving the rising edge of bit4 of 6040_h (0380_h), the slave station makes a judgment on whether it can receive the new displacement instruction: if bit12 of 6041_h (0381_h) is 0 at this time, it indicates that the slave station can receive the new displacement instruction ①; after the slave station receives the new displacement instruction, bit12 of 6041_h (0381_h) is set from 0 to 1, which indicates that the new

	displacement instruction ① has been received and the slave station is currently in the state of not being able to continue to receive the new displacement instruction. After receiving the new displacement instruction, the slave station will set bit12 of 6041 (0381) from 0 to 1, indicating that the new displacement instruction ① has been received, and the current slave station is in the state of not being able to continue to receive new displacement instructions. In the immediate update mode, once a new displacement instruction is received (bit12 of 6041_h (0381_h) changes from 0 to 1), the motor immediately executes the displacement instruction.
(4)	After the host computer receives the bit12 of the slave's status word 6041_h (0381_h) and changes it to 1, it can release the data of the displacement instruction and set bit4 of the control word 6040_h (0380_h) from 1 to 0, indicating that there is no new position instruction.
(5)	When the slave detects that bit4 of the control word 6040_h (0380_h) has changed from 1 to 0, it can set bit12 of the status word 6041_h (0381_h) from 1 to 0, indicating that the slave is ready to receive a new displacement command.

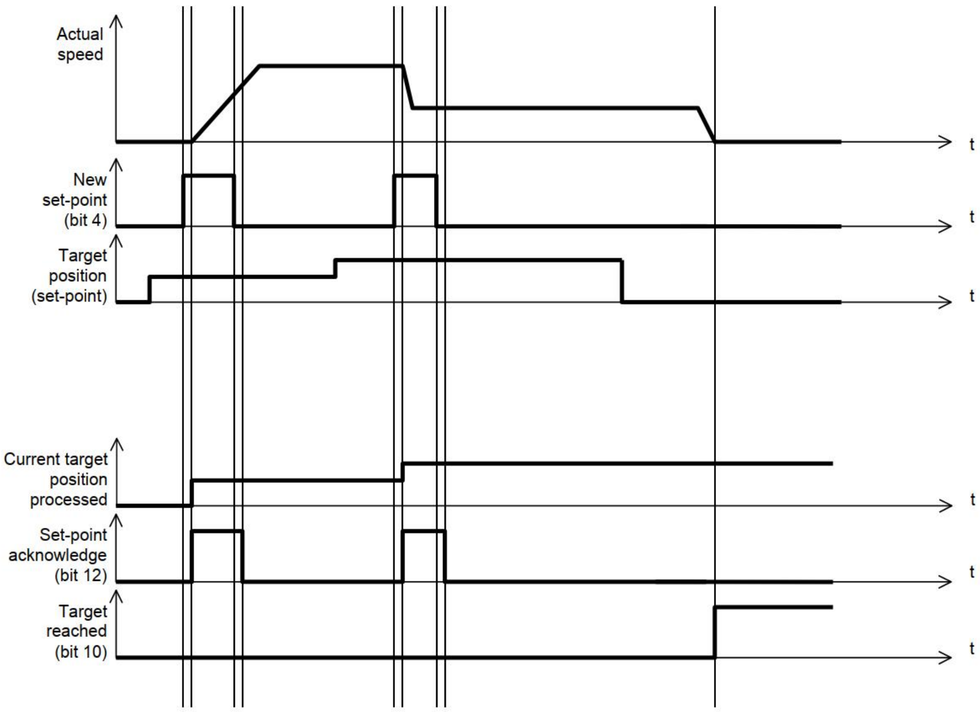

Figure 5- 11

2） Control Instruction Timing - Not Immediately Updated

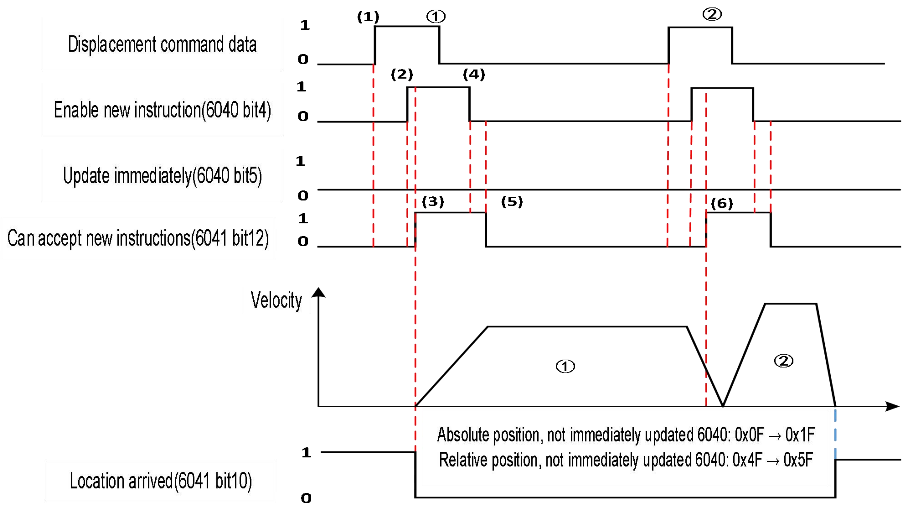

Figure 5- 12

In the case of non-immediate updating, the labels (1)-(5) in the figure are similar to the interpretation in the case of immediate updating, which can be seen by comparing position (6) in the figure. The difference is that: in the case of non-immediate updating, although the new target position (2) is accepted, it is necessary to arrive at position (1) first, decelerate and stop, and then travel to position (2) again.

In the case of non-immediate updates, the motor is designed with 5 caches, which means that there can be 5 target positions in the sequence at the same time, and the motor will travel to these positions in sequence. The motor will travel to these positions in sequence. If the caches are full, it will not be able to accept new positions until one of the caches is empty. These caches are cleared when the motor stops.

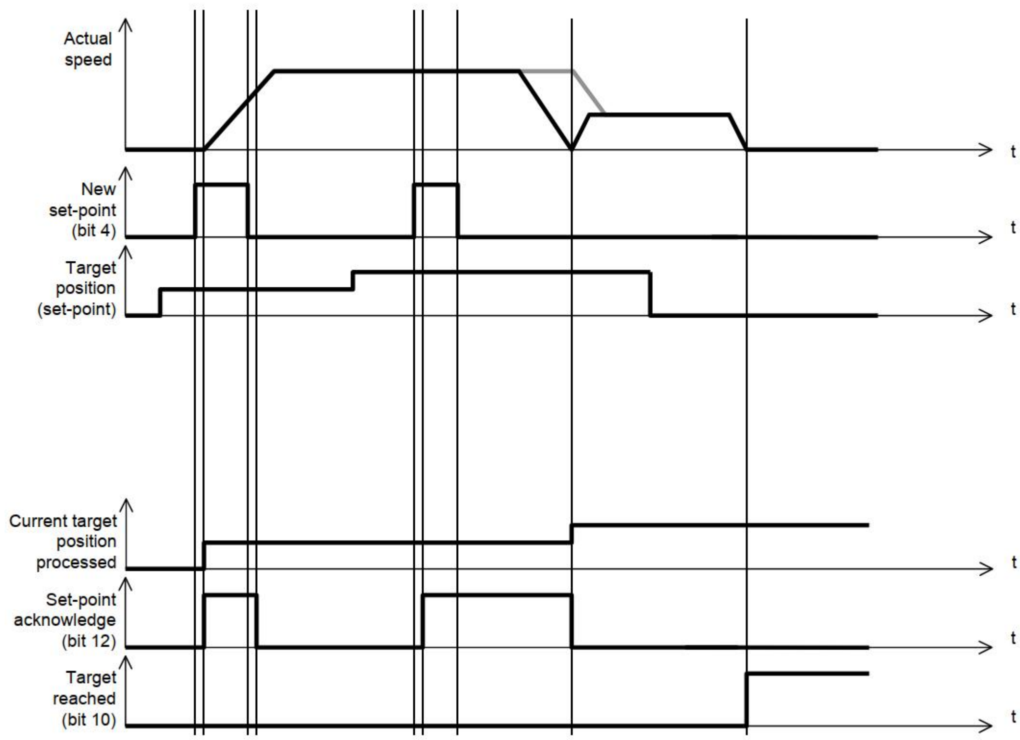

Figure 5- 13

5.3.5 Shock-constrained and Unconstrained Modes

In principle, models can be categorized as "shock-constrained" or "shock-unconstrained" according to shocks.

- **impingement limited model**

Setting object 6086_h (0402_h) to "3" enables the execution of shock-restricted positioning runs. In this case, the shock-related items in object $60A4_h$: 01_h~ 04_h ($041E_h$ ~0424_h) take effect.

- **Impact unconstrained mode**

In this mode, there is no shock limitation at any position of the characteristic curve. To run the "Shock unrestricted" ramp: Set the entry in object 6086_h (0402_h) to "0".

5.3.6 Configuration Example

- Configuration mode:

2002:01_h ($00B1_h$) = 0, Run Mode 6060_h ($03C2_h$) = 0x01 to make the device work in Profile Position Mode;

Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)

Send: 01 10 03 C2 00 01 02 00 01 (set to contour position mode)

- Parameter Configuration:

Write target location $607A_h$ ($03E7_h$) (user unit);

Write Current Segment Displacement Command Uniform Running Speed 6081_h ($03F8_h$) (User Units/s);

Sets the acceleration 6083_h ($03FC_h$) (user units/s^2) and deceleration 6083_h ($03FE_h$) (user units/s^2) of the displacement;

Send: 01 10 03 E7 00 02 04 00 00 27 10 (set target location $607A_h$ ($03E7_h$) to 10000)

Send: 01 10 03 F8 00 02 04 00 00 27 10 (set target speed 6081_h ($03F8_h$) to 10000)

Send: 01 10 03 FC 00 02 04 00 00 9C 40 (sets acceleration 6083_h ($03FC_h$) to 40000)

Send: 01 10 03 FE 00 02 04 00 00 9C 40 (set deceleration 6083_h ($03FE_h$) to 40000)

- Write control word to enable motor

6040_h (0380_h) = 0x06→0x07→ 0x(n)F, motor enable:

Send: 01 10 03 80 00 01 02 00 06 (set 6040_h (0380_h) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040_h (0380_h) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 0F (set 6040_h (0380_h) to 0xF to enable motor)

- Make the motor run

6040_h (0380_h) = 0x(n)F → 0x(n+1)F, motor is running:

The different types of instructions are shown in the table below:

Table 5- 11

6040h (0380h)-bit6	6040h (0380h)-bit5	6040h (0380h) **change**	**Description**
0	0	0x0F → 0x1F	Absolute location, not immediately updated
0	1	0x2F → 0x3F	Absolute location. Update immediately.
1	0	0x4F → 0x5F	Relative position, not immediately updated
1	1	0x6F → 0x7F	Relative position, update immediately

Send: 01 10 03 80 00 01 02 00 5F (set 6040h (0380h) to 0x5F, relative positional motion, not immediate update)

Send: 01 10 03 80 00 01 02 00 1F (set 6040h (0380h) to 0x1F, absolute positional motion, not immediate update)

Send: 01 10 03 80 00 01 02 00 6F (set 6040h (0380h) to 0x6F, relative positional motion, update immediately)

Send: 01 10 03 80 00 01 02 00 3F (set 6040h (0380h) to 0x3F, absolute position motion, update immediately)

- Monitoring Parameters:

  Actual position feedback: 6063h (03C6h) (encoder units), 6064h (03C8h) (user units)

5.4 Velocity Mode (VM)

5.4.1 Structure Plan

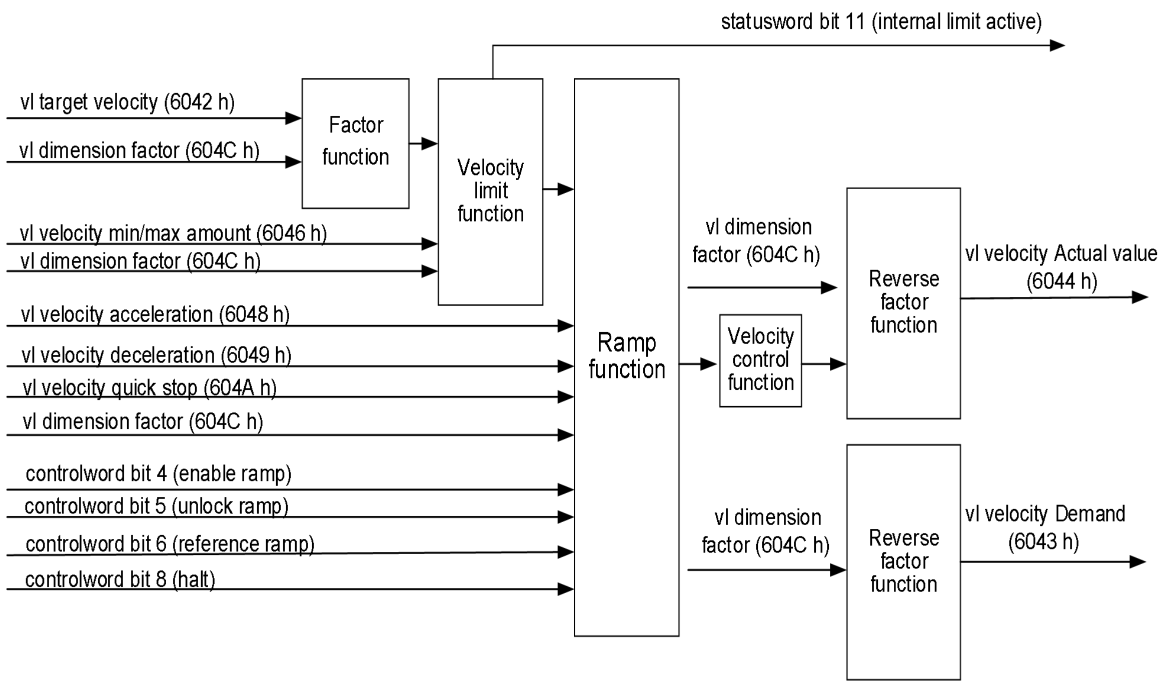

Figure 5- 14

5.4.2 Relevant Object

The following objects require attention in this mode:

Table 5- 12

Object Index	Description
6042_h (0382_h)	Target speed for VM mode (default unit: rpm, related to $604C_h$ (0394_h and 0396_h))
6043_h (0383_h)	Target speed command in effect for VM mode (default unit: rpm)
6044_h (0384)$_h$	Current actual speed feedback value (rpm)
$604C_h$: 01_h (0394_h) and $604C_h$: 02_h (0396)$_h$	Scaling factor for VM mode speed units, default speed units are rpm (revolutions per minute). $604C_h$: 01_h (0394_h) contains the numerator (multiplier) used to calculate speed and $604C_h$: 02_h (0396_h) contains the denominator (divisor). Speed units:rpm* $\frac{604C_h:01_h(0394_h)}{604C_h:02_h(0396_h)}$, for example, if $604C_h$: 01_h (0394_h) is set to 2, and $604C_h$: 02_h (0396_h) is set to 1, the speed unit: 2rpm. If either $604C_h$: 01_h (0394_h) and $604C_h$: 02_h (0396_h) is 0, the speed unit is rpm.
6048_h : 01_h (0389_h) and 6048_h : 02_h ($038B$)$_h$	VM mode acceleration. 6048_h : 01_h (0389_h) contains the speed change (in rpm) and 6048_h : 02_h ($038B_h$) contains the corresponding time (in seconds). The two together calculate the acceleration, the specific formula is as follows: Vl velocity acceleration =Delta speed (6048_h : 01_h (0389_h)) / Delta time (6048_h : 02_h ($038B_h$)) For example, if you need to accelerate the motor to 300rpm in 3.5s, configure 6048_h : 01_h (0389_h) = 3000, 6048_h : 02_h ($038B_h$) = 35.
6049_h : 01_h ($038C_h$) and 6049_h : 02_h ($038E$)$_h$	Deceleration of the VM mode. 6049_h : 01_h ($038C_h$) contains the speed change in rpm, 6049_h : 02_h ($038E_h$) contains the corresponding time in s (seconds) The two together calculate the acceleration as follows: Vl velocity deceleration = Delta speed (6049_h : 01_h ($038C_h$)) / Delta time (6049_h : 02_h ($038E_h$))
$604A_h$: 01_h ($038F_h$) and $604A_h$: 02_h (0391)$_h$	Deceleration speed at fast stop for VM mode. $604A_h$: 01_h ($038F_h$) contains the speed change (in rpm) and $604A_h$: 02_h (0391_h) contains the corresponding time (in s (seconds)). The two together calculate the acceleration as follows: Vl Quick stop deceleration = Delta speed ($604A_h$: 01_h ($038F_h$)) / Delta time ($604A_h$: 02_h (0391_h))
6046_h : 01_h (0385_h) and 6046_h : 02_h (0387)$_h$	Limits for speed in VM mode (default unit: rpm, related to $604C_h$: 01_h (0394_h) and $604C_h$: 02_h (0396_h)) It is described as follows: 6046_h : 01_h (0385_h) Sets the minimum speed. If the target speed

Object Index	Description
	(6042h (0382h)) is lower than the minimum speed, its value will be set to the minimum speed 6046h : 01h (0385h). 6046 h : 02h (0387h) sets the maximum speed. If the target speed (6042h (0382h)) is higher than the maximum speed, its value will be set to the maximum speed 6046 h : 02h (0387h).

The figure below illustrates the role of acceleration, deceleration on the running process:

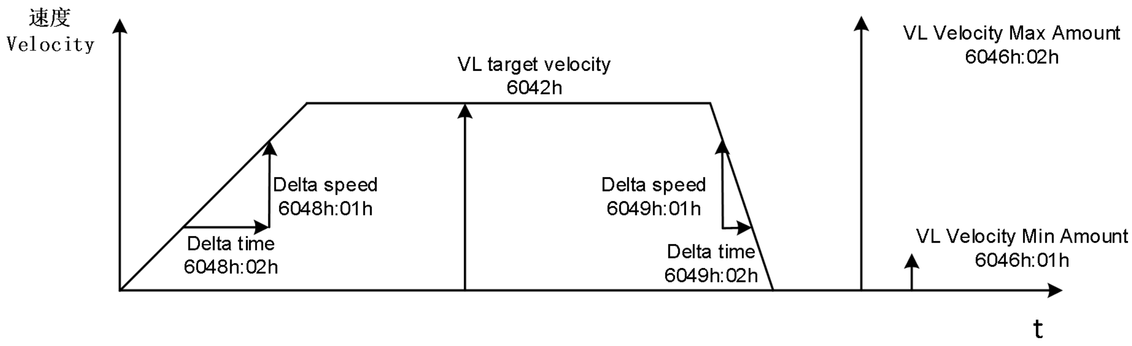

Figure 5- 15

5.4.3 Control Commands and Status Messages

start

To enable this mode, must be set to "0" in object 2002h : 01h (00B1h) and "2" in object 6060h (03C2 h).

control word

15 9	8	7	6	5	4	3 0
(see 3.1.2)	Halt	(see 3.1.2)	Reference ramp	Unlock ramp	Enable ramp	(see 3.1.2)
MSB						LSB

In speed mode, the following bits in object **6040**h (0380h) have a special function

Bit	(be) worth	Description
2	0	When a quick stop is triggered, the motor applies the quick brake according to the quick stop deceleration speed set in object 604Ah . The controller then switches to "Switch on disabled".
	1	no movement
4	0	No acceleration or deceleration process, output changes immediately to the given speed
	1	Speed regulation according to acceleration and

		deceleration settings
5	0	No longer following the output of the planner for speed regulation, the speed output value should be locked to the current speed value
	1	Speed regulation according to the speed output of the planner
6	0	Velocity profile planner input is 0
	1	The speed profile planner input is a given target speed
8	0	no movement
	1	Motor Suspension

Table 5- 13

status word

15 14	13 12	11	10	9 0
(see 3.1.3)	reserved	(see 3.1.3)	reserved	(see 3.1.3)
MSB				LSB

In speed mode, the bits in object **6041**$_h$ (0381$_h$) have no special function; see section 4.1.3 status word for the definition of the general-purpose bits.

5.4.4 Functional Description

In this mode, the motor runs at the target speed preset value, similar to an inverter. Unlike the profile speed mode, this mode does not allow the selection of shock-limited ramps.

5.4.5 Configuration Example

- Configuration mode:
  Write 2002$_h$: 01$_h$ (00B1$_h$) = 0, 6060$_h$ (03C2$_h$) = 0x02, configured for velocity mode (VM)
  Send: 01 10 00 B1 00 01 02 00 00 (set to CIA402 mode)
  Send: 01 10 03 C2 00 01 02 00 02 (set to speed mode)
- Parameter Configuration:
  Write target speed: 6042$_h$ (0382$_h$) = 1000;
  Send: 01 10 03 82 00 01 02 03 E8 (set target speed to 1000 rpm)
  Write acceleration and deceleration:
  6048$_h$: 01$_h$ (0389$_h$) = 500, 6048$_h$: 02$_h$ (038B$_h$) = 1; and
  6049$_h$: 01$_h$ (038C$_h$) = 500, 6049$_h$: 02$_h$ (038E$_h$) = 1;
  Send: 01 10 03 89 00 02 04 00 00 01 F4 (set 6048$_h$: 01$_h$ (0389$_h$) to 500)
  Send: 01 10 03 8B 00 01 02 00 01 (set 6048$_h$: 02$_h$ (038B$_h$) to 1)
  Send: 01 10 03 8C 00 02 04 00 00 01 F4 (set 6049$_h$: 01$_h$ (038C$_h$) to 500)
  Send: 09 10 03 8E 00 01 02 00 01 (set 6049$_h$: 02$_h$ (038E$_h$) to 1)
- Write the word control:
  6040$_h$ (0380$_h$) = 0x06→0x07→ 0x7F, motor enable:

Send: 01 10 03 80 00 01 02 00 06 (set 6040_h (0380_h) to 0x6 to make motor ready)

Send: 01 10 03 80 00 01 02 00 07 (set 6040_h (0380_h) to 0x7 to disable motor)

Send: 01 10 03 80 00 01 02 00 7F (set 6040_h (0380_h) to 0x7F to enable motor)

- Parameter monitoring:

  Current actual speed $606C_h$ ($03D5_h$) (in rpm)

5.5 Profile Velocity Mode (PV)

In this mode, the upper controller sends the target speed, acceleration, and deceleration to the driver, and the speed and torque adjustments are performed internally.

5.5.1 Structure Plan

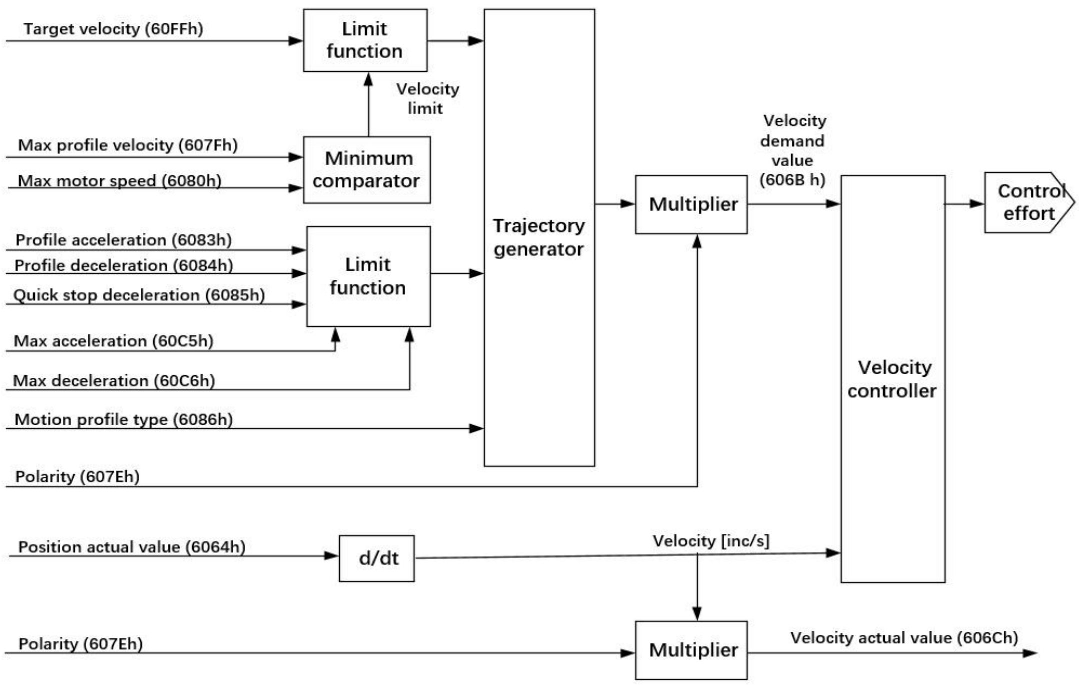

Figure 5- 16

5.5.2 Related Objects

The following objects require attention in this mode:

Table 5- 14

Object Index	Description
6063_h ($03C6_h$)	Feedback of the current absolute position of the motor (encoder unit)
6064_h ($03C8_h$)	Feedback on the current absolute user position of the motor (user units)
$606B_h$ ($03D3_h$)	Speed command value validated within the controller
$606C_h$ ($03D5_h$)	Current actual speed feedback value (rpm)
$607E_h$ ($03F3_h$)	Direction of rotation (polarity), see "4.3 $607E_h$: Polarity".
$607F_h$ ($03F4_h$)	Maximum profile speed during operation (user unit/s), which acts as a speed limiter.
6080_h ($03F6_h$)	Maximum speed of motor (rpm)
6083_h ($03FC_h$)	Profile acceleration during operation (user unit/s^2), size limited by $60C5_h$.
6084_h ($03FE_h$)	Contour deceleration during the run (user unit/s^2), size limited by $60C6_h$.
6085_h (0400_h)	Stopping deceleration rate when performing a "quick stop" (user unit/s^2)
6086_h (0402_h)	Select the type of curve, i.e. slope. If the value is "0", the shock (plus acceleration) is not limited, i.e., the trapezoidal curve; If the value is "3", the shock (plus acceleration) will be limited by the value in $60A4_h$: 01_h -02_h , i.e. the S-curve, only the 01_h and 02_h indexes are used in this device.
$60C5_h$ ($043B_h$)	Maximum acceleration limits (user units/s^2)
$60C6_h$ ($043D_h$)	Maximum deceleration limit (user units/s^2)
$60FF_h$ (0448_h)	Target speed (user units/s)

5.5.3 Control Commands and Status Messages

Enable

To enable this mode, the value "0" must be set in object 2002_h : 01_h (00B1h) and "3" in object 6060_h ($03C2_h$).

control word

15 … 9	8	7	6 … 4	3 … 0
(see 3.1.2)	Halt	(see 3.1.2)	reserved	(see 3.1.2)
MSB				LSB

In profile speed mode, the following bits in object 6040_h (0380_h) have a special function